

## UNIT V

### 5 Types Problem

**Type I,  $[f(x) = e^{ax}]$**

1.  $(D^2 - 4D + 13)y = e^{2x}$ .
2.  $(D^2 - 5D + 6)y = e^{-x} + 3$ .
3.  $(D^5 - D)y = 12e^x$
4.  $(D^2 + 2D + 1)y = e^{-x} + 3$ .
5.  $(D^2 - 4D + 4)y = e^{2x}$ .
6.  $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 3e^{-2x} + e^{-3x}$ .
7.  $y'' + y = 10$ .
8.  $(D^2 - 2D + 1)y = \cos hx$ .
9.  $(D^3 - 1)y = e^{2x}$ .
10.  $(D^2 - 1)y = \sin hx$ .
11.  $\frac{d^2y}{dx^2} - 4y = 3^x$ .
12.  $(D - 2)^2y = e^{2x}$ .
13.  $(D^3 - 3D^2 + 4D - 2)y = e^x$ .
14.  $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{-2x}$
15.  $(D^3 + 8)y = e^{-2x}$ .
16.  $(D^2 - 4)y = \cos h 2x$ .

**Type II  $[f(x) = \sin ax \text{ (or) } \cos ax]$**

1.  $(D^2 + D + 1)y = \sin 2x$ .
2.  $(D^2 - 4D + 3)y = \sin 3x \cos 2x$ .
3.  $(D^2 + 5D + 6)y = \cos^2 x$ .
4.  $(D^2 + 16)y = \cos 4x$ .
5.  $(D^2 + 16)y = e^{-3x} + \cos 4x$ . (Model May 2016)
6.  $(D^2 + 2D + 1)y = \cos x$
7.  $(D^2 + 1)y = \sin x$
8.  $(D^2 - 4D + 3)y = \cos 2x$

9.  $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin 3x$
10.  $(D^2 - 4D + 4)y = e^{2x} + \cos 3x$
11.  $\frac{d^2y}{dx^2} + 4y = \sin 2x$
12.  $(D^2 + 6D + 8)y = e^{-2x} + \cos^2 x$
13.  $(D^3 + 4D)y = \cos 2x$
14.  $(D^2 + 1)y = \sin x \sin 2x$  (Model Apr./May 2016)

**Type III** [ $f(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$ ]

1.  $(D^2 + 2)y = x^2$ .
2.  $(D^2 + 5D + 4)y = x^2 + 7x + 9$ .
3.  $(D^4 - 2D^3 + D^2)y = x^2 + e^{-x}$ .
4.  $(D^4 + 8D^2 + 16)y = 16x + 10$ .
5.  $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = x^2 + 3$ .
6.  $(D^3 - D^2 - 6D)y = x^2 + 1$ .
7.  $(D^4 - 2D^3 + D^2)y = x^3$ .
8.  $(D^4 + 8D^2 + 16)y = 16x + 10$ .
9.  $(D^3 - 3D^2 - 6D + 8)y = x$ .
10.  $(D^2 + 1)y = x^3$ .

**Type IV** [ $f(x) = e^{ax} X$ ]

1.  $(D^3 - 3D^2 + 3D - 1)y = x^2e^x$ .
2.  $(D^2 - 4D + 13)y = e^{2x} \cos 3x$ .
3.  $(D^2 + 4D + 3)y = e^{-x} \sin x + xe^{3x}$ .
4.  $(D^2 + 2D - 3)y = e^{2x}(1 + x^2)$ . (Model May 2015)
5.  $(D^2 + 2D - 3)y = e^x \cos x + e^{-x}x^2$ .
6.  $(D^2 + 2D + 5)y = e^x \sin 2x + x$ .
7.  $(D^2 + 4D + 4)y = e^{-x} \sin 2x$ .
8.  $(D^2 + 9)y = (x^2 + 1)e^{3x}$ .
9.  $(D^2 - 2D - 1)y = x^2e^{3x}$ .

10.  $(D^2 + 4D + 3)y = e^x \cos 2x - \cos 3x$ .

**Type V [ $f(x)x^n \sin ax$  (or)  $x^n \cos ax$ ]**

1.  $(D^2 + 4)y = x \sin x$ .
2.  $(D^2 - 1)y = x^2 \cos x$ .
3.  $(D^2 - 2D + 1)y = xe^x \sin x$ .
4.  $(D^2 - 2D + 1)y = x \sin x$ .
5.  $\frac{d^2y}{dx^2} - y = x \sin x + e^x + x^2 e^x$
6.  $(D^4 + 2D^2 + 1)y = x^2 \cos x$
7.  $(D^2 - 1)y = x^2 \cos x$
8.  $(D^2 + 16)y = x^2 \sin 2x$

**❖ Method of variation of Parameters**

1.  $\frac{d^2y}{dx^2} + 4x \frac{dy}{dx} = 4 \tan 2x$ . (Nov. 2016)
2.  $(D^2 + 1)y = \tan x$ . (Jan. 2014)
3.  $(D^2 + 1)y = \sec x$ .
4.  $(D^2 + 1)y = \operatorname{cosec} x$ . (May 2017, Dec. 2017)
5.  $(D^2 - 2D + 2)y = e^x \tan x$ .
6.  $(D^2 + 2D + 5)y = e^{-x} \tan x$ .

**❖ Euler's Type**

1.  $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = \sin(\log x)$ . (May 2017)
2.  $x^2 \frac{d^2y}{dx^2} - 7x \frac{dy}{dx} + 12y = x^2$ .
3.  $\frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$ .
4.  $[x^2 D^2 - 2xD + 4]y = x^2 + 2 \log x$
5.  $[x^3 D^3 + 3x^2 D^2 + xD + 1]y = \sin(\log x)$ .
6.  $[x^2 D^2 - xD + 2]y = x \log x$ . (Model May 2015)

**❖ Simultaneous Equation**

1.  $\frac{dx}{dt} - \frac{dy}{dt} - y = e^{-t}; \frac{dy}{dt} + x - y = e^{2t}$

2.  $\frac{dx}{dt} + 2x - 3y = t; \frac{dy}{dt} - 3x + 2y = e^{2t}$ . (Model May 2017, Nov. 2016)
3.  $\frac{dx}{dt} - \frac{dy}{dt} + 2y = \cos 2t; \frac{dx}{dt} + \frac{dy}{dt} - 2x = \sin 2t$ .
4.  $\frac{dx}{dt} + y = \sin t; \frac{dy}{dt} + x = \cos t$  given that  $x = 2$  &  $y = 0$  at  $t = 0$ .
5.  $\frac{dx}{dt} + 2y = -\sin t; \frac{dy}{dt} - 2x = \cos 2t$
6.  $\frac{d^2x}{dt^2} + y = \sin t; \frac{d^2y}{dt^2} + x = \cos t$  (May 2015)
7.  $\frac{dx}{dt} - y = 0; \frac{dy}{dt} + x = 0$ .
8.  $\frac{dx}{dt} + y = e^t; x - \frac{dy}{dt} = t$ .
9.  $\frac{dx}{dt} + 2y + \sin t = 0; \frac{dy}{dt} - 2x - \cos t = 0$ .
10.  $\frac{dx}{dt} + 5x - 2y = t; \frac{dy}{dt} + 2x + 5y = 0$  so that  $x = y = 0$  when  $t = 0$ .